

CN -- Computer Networks Study Notes Index

Format: Definition + Mnemonic (Hinglish) + Example + 40s script + Follow-ups.

Revision Order

- Step 1 01-DATA-COMM-AND-BASICS.md 5 components + 3 data flow modes (10 min)
- Step 2 02-OSI-MODEL.md 7 layers -- HIGH PRIORITY (15 min)
- Step 3 03-NETWORK-TOPOLOGY.md 6 topologies + comparison table (12 min)

Day-before quick revision: Cheat sheet at bottom of each file only.

OSI model is #1 priority -- say the mnemonic aloud 3 times right now.

Files

File

Topics

Read Time

Status

01-DATA-COMM-AND-BASICS.md

5 DC components, Simplex/Half/Full Duplex, Network criteria, Point-to-point vs Multipoint

8 min

[OK] Done

02-OSI-MODEL.md

All 7 layers (function, protocol, device, data unit), OSI vs TCP/IP

12 min

[OK] Done

03-NETWORK-TOPOLOGY.md

6 topologies with full comparison (advantage, disadvantage, reliability, cost, links formula)

12 min

[OK] Done

Master Cheat Sheet

Data Communication:

5 components MSRTTP "Mere Saath Raho Tum Pyaar se"
Message, Sender, Receiver, Transmission Medium, Protocol

Data Flow:

Simplex Sirf ek taraf (radio, keyboard)

Half Duplex Baari baari (walkie-talkie)

Full Duplex Ek saath dono (phone call)

OSI 7 Layers (Bottom Top):

"Please Do Not Throw Sausage Pizza Away"

Physical Data Link Network Transport Session Presentation Application

Data Units:

Physical=Bits, Data Link=Frame, Network=Packet, Transport=Segment, Upper 3=Data

Key Layer Facts:

Layer 1 Physical Hub, raw bits, RJ-45

Layer 2 Data Link Switch, Frame, MAC address, ARP

Layer 3 Network Router, Packet, IP address, ICMP

Layer 4 Transport Firewall, Segment, TCP/UDP

Layer 7 Application HTTP, FTP, SMTP, Telnet

TCP vs UDP:

TCP reliable, connection-oriented, slower (email, web, file)

UDP unreliable, connectionless, faster (video, DNS, gaming)

Topologies (6): "Bahut Star Ring Tree Mesh Hybrid hai"

Bus single cable, cheap, backbone fail = all down

Star central hub, easy, hub fail = all down, max 100m

Ring unidirectional, one node fail = all down

Tree hierarchical star, root fail = all down

Mesh $n(n-1)/2$ links, most reliable, most expensive

Hybrid combination, most complex, worst security

Topology quick pick:

Most reliable Mesh

Most common office Star

Most scalable Tree

Most complex Hybrid

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Topics To Add (Next Pages)

[] TCP/IP protocol stack detail

[] IP addressing -- IPv4, subnetting, CIDR

[] DNS -- resolution process (step-by-step)

[] HTTP vs HTTPS

[] Error detection/correction (CRC, parity)

[] Routing protocols (RIP, OSPF, BGP)